TEXORA · LECTURE SHEET

# Production Studies, Machine Utilisation, HOK and Machine Interference

*Measuring what a running mill actually achieves — production studies of machine utilisation and operator performance, the HOK labour-intensity measure of spinning, and the machine-interference problem that limits how many machines one operator can tend.*

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 5 · LEVEL 4 TERM 1

## 1. Production studies

A **production study** is a long-duration measurement of a machine, section or operator over a full shift or longer, recording running time, stoppages by cause, and output. Unlike a time study (minutes, one operation), a production study answers departmental questions: What fraction of the shift do machines actually produce? What stops them, for how long, and whose problem is each stop? Stoppages are coded — material, mechanical, electrical, quality, operator, waiting — because an uncoded "idle" total cannot be acted on. Modern mills collect the same data automatically from machine monitoring systems; the logic is the production study's logic, only the clipboard has changed.

## 2. Machine utilisation and efficiency

Utilisation % = Running time ÷ Available time × 100 | Lost time is analysed by cause, never as one lump

**WORKED EXAMPLE — LOOM PRODUCTION STUDY (ONE SHIFT, 480 MIN)**

Running (producing cloth): **430 min**. Stops: warp breaks 20 min, weft breaks 15 min, mechanical repair 15 min — total stopped 50 min.

Utilisation = 430 ÷ 480 × 100 = **89.6 %**.

Reading by cause: warp breaks (20 min) point to warp preparation/sizing quality; weft breaks (15 min) to weft package quality or loom settings; mechanical (15 min) to maintenance. Three different owners for three stoppages of similar size — the coded study, not the 89.6 % headline, is the management product.

Utilisation must be distinguished from **efficiency**: utilisation asks how much of the available time the machine ran; efficiency compares the output during running with the output the machine should have given at its rated speed (a loom running all shift at reduced speed shows 100 % utilisation and poor efficiency). Both numbers are needed; either alone can flatter a badly run shed.

## 3. Operator performance

Operator performance is measured against the standard times of Topic 3:

$$\text{Performance \%} = (\text{ Standard time} \times \text{Quantity produced }) \div \text{Time taken} \times 100$$

An operator producing 440 pieces in 480 min on an operation of SMV 1.00 min has earned 440 standard minutes: performance = 440 ÷ 480 × 100 ≈ 91.7 %. Performance figures feed incentive schemes and identify training needs — and, used fairly, they also expose bad standards: if every operator on an operation sits at 130 %, the standard, not the workforce, is wrong.

## 4. HOK — operator hours per 100 kg of production

In spinning (and by extension other mass-production sections) labour intensity is tracked as **HOK — operator hours per 100 kg of production** (from the German *Stunden pro 100 kg* tradition of spinning-mill cost comparison). It collapses staffing and output into one comparable number:

$$\text{HOK} = \text{Total operator-hours worked} \div \text{Production (kg)} \times 100$$

**WORKED EXAMPLE — HOK OF A WEAVING SHED (ONE WEEK)**

24 weavers work 48 h each in the week: operator-hours = 24 × 48 = 1,152 h.
Production in the same week: 4,800 kg of fabric.
HOK = 1,152 ÷ 4,800 × 100 = **24.0 operator-hours per 100 kg**.
Use: compare across weeks, sheds and mills on the same product mix. A rising HOK with unchanged product means labour is growing faster than output — more operators per machine, more stoppages being attended, or falling machine utilisation quietly inflating the numerator's work content.

HOK's discipline is that it cannot be improved by reclassifying output or by overtime accounting tricks: hours are hours, kilograms are kilograms. Its limit is product sensitivity — fine counts and complex fabrics legitimately need more operator attention — so comparisons must be like-for-like.

## 5. Machine interference

When one operator tends several machines that stop at random (looms, ring frames, winders), a machine that stops while the operator is busy at another must **wait**. This waiting is **machine interference**: production lost not because machines are slow or operators idle in total, but because random stoppages collide. Interference grows steeply with the number of machines per operator and with the ratio of attention time to running time.

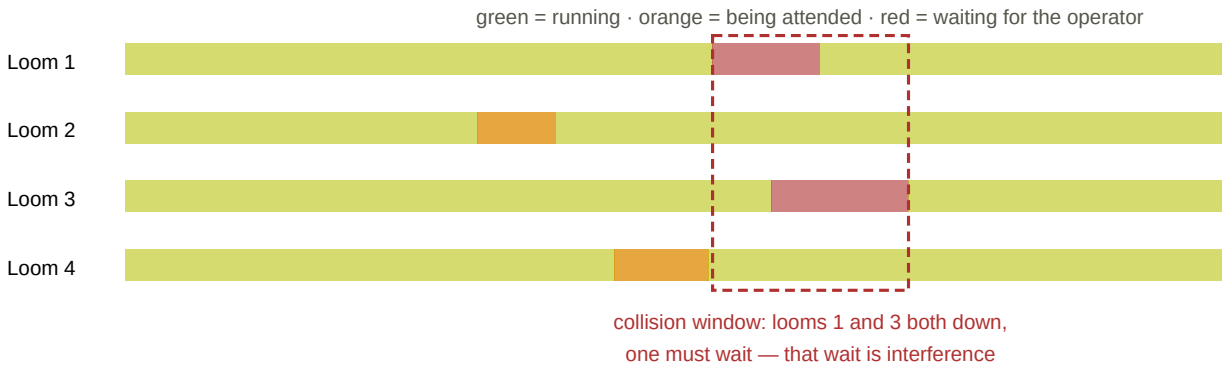



Fig. 1 — Machine interference: random stoppages overlap, and one machine waits while the operator attends another.

**Sizing the problem.** Suppose each loom runs on average 54 min per hour and needs 6 min of operator attention per hour. With 6 looms, total attention demand = 6 × 6 = 36 min per hour — 60 % of one operator's time. Average load is feasible; the loss comes from coincidence: when two looms stop together, one waits its turn. Practical consequences: (i) interference loss rises faster than proportionally as machines per operator increase — going from 4 to 8 looms per weaver much more than doubles waiting; (ii) reducing attention time per stop (better packages, quicker piecing) cuts interference twice over — less waiting and less service; (iii) the economic optimum assignment balances operator idle time against machine waiting time, and it is found by production study and interference tables/queueing models, not by guessing. Where interference is allowed for in standards, it enters as an **interference allowance** on the machine-paced output.

## Glossary

**Production study** — long-duration record of running time, stoppages by cause, and output. **Utilisation** — running time as a percentage of available time. **Efficiency (machine)** — actual output while running vs rated-speed output. **Operator performance** — standard minutes earned ÷ minutes taken × 100. **HOK** — operator-hours per 100 kg of production; a labour-intensity benchmark. **Machine interference** — machine waiting time caused by random stoppages colliding when one operator tends several machines. **Interference allowance** — allowance added to standards for expected interference loss.

## Key points

• Production studies code stoppages by cause — the causes, not the idle total, are actionable.
• Worked study: 430/480 min running = 89.6 % utilisation, split into warp, weft and mechanical causes.
• HOK = operator-hours ÷ kg × 100: 1,152 h and 4,800 kg give HOK 24.0 — compare only like-for-like products.
• Interference: average attention load can be feasible (36 of 60 min) while random collisions still stop machines; it grows steeply with machines per operator.
• Utilisation and efficiency are different numbers; a shed needs both.

## Practice

1. A ring-frame department works 96 operator-hours in a day and produces 350 kg of yarn. Compute HOK. The same department a month later, with the same hours, produces 320 kg. What happened to HOK and what are two possible causes?
2. In a 480-min shift a knitting machine runs 408 min; stops: yarn breaks 30 min, needle damage repair 25 min, waiting for yarn 17 min. Compute utilisation and rank the causes.
3. An operator completes 510 pieces in 480 min on an operation with SMV 0.95 min. Compute performance %.
4. Explain why doubling machines per operator more than doubles interference loss, referring to randomness.

### Answer key

1. HOK = 96 ÷ 350 × 100 ≈ 27.4 operator-hours per 100 kg. Later: 96 ÷ 320 × 100 = 30.0 — HOK rose (worse labour intensity) by about 9 %. Possible causes: more stoppages needing attention (material quality fell), lower machine utilisation spreading the same attendance over less output, a finer count being spun (legitimately more attention — check product mix before judging), or extra operators added without output rising.
2. Stopped total = 30 + 25 + 17 = 72 min; utilisation = 408 ÷ 480 × 100 = 85.0 %. Ranking: yarn breaks 30 (material/package quality), needle repair 25 (maintenance/machine condition), waiting for yarn 17 (logistics). Each has a different owner.
3. Standard minutes earned = 510 × 0.95 = 484.5. Performance = 484.5 ÷ 480 × 100 ≈ 100.9 %.
4. Stoppages are random, so the chance that two or more machines are down simultaneously grows combinatorially with the number tended: with few machines, collisions are rare and the operator (or the machines) wait little; with many, at busy moments several machines queue for one pair of hands while at quiet moments the operator is idle — average load is unchanged in proportion, but waiting concentrates in the collisions. The loss therefore rises faster than the assignment, which is why interference tables show sharply increasing loss percentages as machines per operator grow, especially when attention time is a large fraction of running time.